

Integrating Urban Soil Microbiome Profiling with Machine Learning-Guided Culture Medium Optimization

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Abstract

Urban soils harbor diverse bacterial communities that are shaped by local environmental conditions and land-use characteristics. However, microbial differences between urban roadside and park soils remain insufficiently characterized, particularly in Korean metropolitan environments, and the efficient cultivation of soil-derived bacteria remains challenging because of the large experimental space associated with medium optimization. In this study, we characterized bacterial communities in 32 soil samples collected from roadside and park sites across the Seoul–Gyeonggi metropolitan region of the Republic of Korea and explored a machine learning-guided strategy for optimizing the culture conditions of a soil-derived bacterial isolate.

Bacterial community composition was assessed by 16S rRNA gene amplicon sequencing of the V4 region. Overall comparisons of roadside and park soils revealed substantial overlap in alpha- and beta-diversity patterns. Exploratory LEfSe analysis at the order level identified Microtrichales as a discriminative feature under the predefined criteria (FDR-adjusted $P = 0.083$; LDA score = 3.8). Nevertheless, linear discriminant analysis effect size (LEfSe) analysis at the order level identified several bacterial orders with discriminative effect sizes between the two land-use groups ($|LDA| \geq 2.0$). A post hoc exploratory analysis of microbiologically contrasting sample subsets showed more pronounced differences in community composition and diversity; however, these patterns were considered descriptive because subset selection was based on the original microbiome profiles.

Culturable bacterial strains were subsequently isolated from the collected soils, and *Priestia megaterium*, independently obtained from both roadside and park samples, was selected for culture-medium optimization. Following an initial screening of 12 medium compositions, a METIS-guided experimental round was conducted using peptone, meat extract, yeast extract, and glucose as optimization variables and OD600 as the growth response. An XGBoost-based

ensemble model identified meat extract as the most influential predictor of growth (0.360 ± 0.076), followed by yeast extract (0.276 ± 0.049), glucose (0.253 ± 0.081), and peptone (0.111 ± 0.114). Active learning generated four candidate compositions for subsequent testing, converging on 5 g/L peptone, 9.60–10.88 g/L meat extract, 0.64–3.12 g/L yeast extract, and 60 g/L glucose. Pairwise interaction analysis further indicated strong context-dependent effects involving peptone, particularly with meat extract and glucose.

Together, these findings provide an exploratory characterization of bacterial communities in Seoul–Gyeonggi urban soils and demonstrate the feasibility of integrating microbiome profiling with machine learning-guided culture optimization. This framework provides a basis for future studies incorporating independently defined environmental variables, functional characterization of soil-derived isolates, and experimental validation of model-recommended culture conditions.

Keywords Urban soils, Biodiversity, Bacterial community, 16S rRNA sequencing, Active learning, Machine learning, METIS

Introduction

The microbiome encompasses microbial communities residing within a specific habitat together with their genetic and functional potential. Microbial communities are distributed across a wide range of environments, including the human body, soil, water, and air, where they contribute to diverse ecological and biological processes [1]. In humans, microbiome composition varies substantially among body sites and is closely associated with host physiology and health [2]. More recently, increasing attention has been directed toward the interface between human-associated and environmental microbiomes, as humans are continuously exposed to environmental microorganisms through contact with soil and vegetation and through inhalation of airborne particles and dust [3]. Studies examining exposure to green spaces have reported shifts in skin and respiratory microbial communities toward profiles resembling those of surrounding soil or airborne microbiota, suggesting that local environmental microbial communities may influence patterns of human microbial exposure [3]. Although such exposure does not necessarily indicate stable colonization, these findings highlight the importance of characterizing environmental microbiomes in settings that are frequently encountered by urban populations.

Soil is one of the most diverse microbial reservoirs on Earth and plays fundamental roles in nutrient cycling, decomposition, carbon dynamics, and overall ecosystem functioning [4]. Urban soils are particularly complex because they are exposed to intensive anthropogenic disturbance and highly heterogeneous environmental conditions. Their microbial communities can be influenced by land use, vegetation, soil physicochemical properties, pollution, and the intensity of human activity [5]. Despite the ecological importance of urban soils, microbial variation within metropolitan environments remains less well characterized than that of natural or agricultural soils, particularly in rapidly urbanized regions [6].

Roadside and park soils represent two contrasting urban environments. Roadside soils are

continuously exposed to traffic-related disturbances and may accumulate vehicle-derived contaminants, including heavy metals, petroleum-related compounds, and polycyclic aromatic hydrocarbons [7,8]. Such environmental pressures have been associated with changes in soil microbial composition and diversity [9]. In contrast, urban park soils are more strongly influenced by vegetation, organic inputs, and green-space management and generally experience lower direct exposure to vehicular activity [10]. Consequently, comparing bacterial communities between roadside and park soils may provide insight into how different urban land-use environments are associated with microbial community structure. However, considerable heterogeneity may also exist within each land-use category because local environmental conditions can vary substantially among sampling sites. Therefore, community-level comparisons should consider both broad land-use patterns and within-group microbial variability.

High-throughput 16S rRNA gene sequencing provides a culture-independent approach for characterizing bacterial community composition and diversity across complex soil environments [11]. Nevertheless, community profiling alone provides limited information regarding the practical cultivation and manipulation of soil-derived microorganisms. Culture-based isolation remains valuable for obtaining viable bacterial strains that can subsequently be examined under controlled experimental conditions. However, optimization of microbial cultivation is often challenging because growth can depend simultaneously on multiple medium components and their interactions. Conventional trial-and-error approaches become increasingly inefficient as the number of candidate concentrations and combinations expands. Machine-learning-guided experimental design offers an alternative strategy for navigating such multidimensional experimental spaces. Active learning frameworks iteratively use available experimental observations to predict promising combinations while selecting new conditions that balance exploitation of high-performing regions with exploration of uncertain regions [12].

METIS, a machine-learning-guided experimental design framework, was developed to reduce the number of experiments required for optimization of complex biological systems by iteratively integrating experimental measurements with predictive modeling [12]. Such approaches may therefore be useful for optimizing the cultivation of environmental bacterial isolates while minimizing the experimental burden associated with exhaustive medium screening.

In the present study, we characterized bacterial communities in urban roadside and park soils collected across the Seoul–Gyeonggi metropolitan region of the Republic of Korea using 16S rRNA gene sequencing. Taxonomic composition and alpha- and beta-diversity were compared between the two land-use environments, and bacterial taxa showing discriminative patterns were explored. Because substantial heterogeneity was observed within the overall microbiome profiles, an additional post hoc exploratory analysis was performed using microbiologically contrasting sample subsets to further examine community-level variation. In parallel, culturable bacterial strains were isolated from the collected soils, and *Priestia megaterium*, independently recovered from both roadside and park samples, was selected as a model soil-derived isolate for culture optimization. Using experimentally measured growth across different combinations of peptone, meat extract, yeast extract, and glucose, we applied a METIS-based active machine-learning framework to evaluate medium-component contributions and generate candidate conditions for subsequent experimental testing. Through this integrated approach, this study aimed to link urban soil microbiome characterization with data-driven cultivation of a soil-derived bacterial resource and to provide an exploratory framework for future functional and environmental microbiology studies.

Materials and Methods

Study Area and Sampling Site Selection

This study was conducted in the Seoul–Gyeonggi metropolitan region of the Republic of Korea. A total of 32 urban soil samples were collected from roadside and park environments in July 2026. Sampling sites were distributed across Seoul and Gyeonggi Province and were classified into two land-use categories according to their surrounding environmental characteristics. Roadside sampling sites were selected along major roads with relatively high vehicular traffic, and sampling points were located within 1 m of the curb to represent soil environments directly influenced by road traffic. Park sampling sites were selected from designated urban green spaces with minimal direct exposure to vehicular traffic. The two site categories were used to investigate differences in bacterial community composition and diversity associated with contrasting urban land-use environments. The Seoul–Gyeonggi metropolitan region experiences a warm and humid summer climate, with frequent rainfall during the sampling season. The mean summer temperature and precipitation in the region are approximately 26.1 °C and 892.1 mm, respectively [13].

Soil Sampling and Sample Processing

At each sampling site, visible surface debris, including leaves, stones, and other foreign materials, was removed prior to soil collection. Topsoil was collected from a depth of approximately 2–5 cm using sterile sampling tools. Soil from each sampling site was collected into a sterile 15 mL conical tube until the tube was filled to capacity. The collected samples were immediately transported to the laboratory on ice. Upon arrival, soil samples were divided for culture-based bacterial isolation and molecular analysis. Samples designated for bacterial cultivation were processed within 24 h of collection, whereas aliquots

intended for microbial DNA extraction were stored at -80°C until further analysis. Sampling tools were disinfected with 96% ethanol between sampling sites to minimize cross-contamination, and disposable gloves were changed between samples.

DNA Extraction and 16S rRNA Gene Sequencing

Total microbial DNA was extracted from soil samples using the DNeasy PowerSoil Pro Kit (QIAGEN, Hilden, Germany) according to the manufacturer's instructions. The concentration of extracted DNA was quantified using a Qubit 4 Fluorometer (Thermo Fisher Scientific, Waltham, MA, USA), and DNA integrity was assessed by agarose gel electrophoresis. Purified DNA samples were stored at -20°C until further analysis.

Amplicon libraries targeting the V4 region of the bacterial 16S rRNA gene were prepared using the primer pair 515F (5'-GTGCCAGCMGCCGCGGTAA-3') and 806R (5'-GGACTACHVGGGTWTCTAAT-3') [11]. PCR amplification was performed using 2× KAPA HiFi HotStart ReadyMix (Kapa Biosystems, Wilmington, MA, USA) under optimized cycling conditions. The resulting amplicons were purified using AMPure XP beads (Beckman Coulter, Brea, CA, USA) to remove residual primers and small DNA fragments. Purified amplicon libraries were quantified, normalized, and pooled at equimolar concentrations prior to sequencing. Sequencing was performed on an Illumina iSeq 100 platform (Illumina, San Diego, CA, USA) according to the manufacturer's standard protocol, generating single-end reads for downstream microbiome analysis.

Bioinformatic and Microbial Community Analysis

Raw 16S rRNA gene sequencing data were processed using QIIME 2 version 2023.9 [14]. A manifest file was generated in TSV format, and single-end sequence reads were imported as SampleData[SequencesWithQuality] using the SingleEndFastqManifestPhred33V2 format.

Sequence quality control, denoising, and amplicon sequence variant (ASV) inference were performed using the DADA2 plugin [15], with trim-left set to 0 and trunc-len set to 240. Taxonomic assignment was performed using a Naive Bayes classifier trained on the SILVA 99% reference database specific to the 515F/806R region [16]. Taxonomic composition was summarized at multiple taxonomic levels, including phylum, class, order, family, and genus. Microbial diversity was evaluated using both alpha- and beta-diversity analyses. Alpha diversity was assessed using observed features, Chao1, ACE, Shannon, Simpson, and Fisher indices. Differences in alpha-diversity indices between roadside and park soil samples were evaluated using the Mann–Whitney U test. A two-sided P value < 0.05 was considered statistically significant.

Beta diversity was evaluated using Bray–Curtis distance, Jensen–Shannon divergence, and Jaccard distance and visualized by principal coordinates analysis (PCoA). Differences in overall bacterial community composition between roadside and park soils were assessed using permutational multivariate analysis of variance (PERMANOVA) [17].

Taxonomic composition between roadside and park soils was compared descriptively at multiple taxonomic levels. Linear discriminant analysis effect size (LEfSe) analysis was additionally performed at the order level to identify bacterial taxa exhibiting discriminative abundance patterns between the two environmental groups [18]. For LEfSe analysis, an FDR-adjusted P value < 0.10 and a log LDA score ≥ 2.0 were used as the predefined criteria.

To further explore the heterogeneity observed within the complete dataset, an additional post hoc subset analysis was conducted. Samples exhibiting contrasting microbial community profiles in the initial NGS analysis were selected and re-evaluated for taxonomic composition and alpha- and beta-diversity. Because subset selection was based on the same microbiome dataset used for subsequent analyses, this analysis was considered exploratory and descriptive rather than an independent validation of differences between environmental groups.

Isolation and identification of culturable bacteria

Selected soil samples were subjected to culture-based bacterial isolation to recover cultivable bacterial strains. Soil samples were suspended and thoroughly mixed in sterile phosphate-buffered saline (PBS) to obtain homogeneous microbial suspensions. The suspensions were serially diluted when necessary and inoculated onto de Man, Rogosa and Sharpe (MRS) agar and nutrient agar plates. The plates were incubated aerobically at 37 °C for 24 h. Following incubation, morphologically distinct colonies were selected and repeatedly subcultured on fresh agar plates to obtain pure bacterial isolates.

Individual colonies recovered from MRS agar were transferred to MRS broth, whereas colonies recovered from nutrient agar were transferred to nutrient broth and incubated under the same conditions. Genomic DNA was extracted from each purified isolate, and the 16S rRNA gene was amplified and sequenced. The resulting sequences were compared with reference sequences in the NCBI nucleotide database to determine the taxonomic identities of the isolates. Among the recovered isolates, *Priestia megaterium* was selected as a representative soil-derived bacterial isolate for subsequent culture-medium optimization experiments.

Initial medium composition screening

Four medium components—peptone, meat extract, yeast extract, and glucose—were selected as variables for the initial culture-medium screening of *Priestia megaterium*. A total of 12 medium compositions were manually designed to explore a range of concentrations and combinations of the four components prior to machine learning-guided optimization. The tested concentration ranges were 5–20 g/L for peptone, 4–16 g/L for meat extract, 2–8 g/L for yeast extract, and 10–40 g/L for glucose.

The initial experimental design included conditions in which all four components were

simultaneously adjusted to different concentration levels, as well as conditions in which individual components were varied relative to the baseline composition. This design was intended to provide an initial experimental dataset spanning a broad region of the medium-composition space rather than to identify an optimal formulation through exhaustive screening. *P. megaterium* was cultured under each of the 12 medium conditions, and bacterial growth was quantified by measuring optical density at 600 nm (OD600). The resulting OD600 measurements were used as the response variable, whereas the concentrations of peptone, meat extract, yeast extract, and glucose were used as input variables for subsequent machine learning-guided medium optimization.

Machine Learning-Guided Culture-Medium Optimization

Following the initial screening of 12 manually designed medium compositions, the experimental growth data were used as the starting dataset for machine learning-guided culture-medium optimization. Optimization was performed using an active learning framework adapted from METIS (Machine-learning Guided Experimental Trials for Improvement of Systems) [12]. The concentrations of peptone, meat extract, yeast extract, and glucose were used as input variables, whereas experimentally measured OD600 was used as the response variable representing bacterial growth.

Based on the initial experimental dataset, METIS was used to generate a second set of candidate medium compositions. These recommended compositions were subsequently evaluated experimentally, and the resulting OD600 measurements were used for further model development.

An XGBoost regression model (XGBRegressor; objective = reg:squarederror) [19] was used to model the relationship between medium composition and bacterial growth. Model hyperparameters were optimized using RandomizedSearchCV [20], in which 200 randomly

sampled parameter combinations were evaluated using four-fold cross-validation with negative mean absolute error as the scoring criterion. The 20 highest-performing XGBoost models were independently fitted to the experimental dataset and combined to construct an ensemble model. For each candidate medium composition, the predicted growth response was calculated as the mean prediction across the 20 ensemble models.

The medium-composition search space was defined according to experimentally feasible concentration and liquid-handling constraints. Peptone concentrations were selected from 1, 5, 10, and 20 g/L, whereas meat extract and yeast extract were allowed to vary within ranges of 0–16 and 0–8 g/L, respectively. Glucose concentrations were explored within the predefined concentration space used in the METIS workflow. The final reaction volume was set to 1 mL, with 25% of the total volume reserved as a fixed component. A minimum dispensing volume of 1 μ L and a maximum model-controlled volume of 750 μ L were applied to exclude experimentally infeasible combinations.

Active learning was performed using the Bayesian optimization procedure implemented in the METIS framework. Candidate compositions were selected by balancing exploitation of regions predicted to support high bacterial growth with exploration of less-characterized regions of the medium-composition space. A pool of 10,000 feasible candidate compositions was generated, from which four informative compositions were selected as recommendations for a subsequent experimental round.

In addition, 5,000 feasible medium compositions were randomly sampled from the predefined search space and evaluated using the trained ensemble model. Candidate compositions were ranked according to their ensemble-predicted OD600 values to identify regions associated with high predicted bacterial growth. Because the final model-recommended compositions were not experimentally evaluated in an additional cultivation round, they were interpreted as predicted candidate formulations rather than experimentally confirmed optimal media.

Feature importance and interaction analysis

The relative contribution of each medium component to growth prediction was evaluated using feature importance scores obtained from the XGBoost ensemble model. Feature importance values for peptone, meat extract, yeast extract, and glucose were extracted from each of the 20 highest-performing XGBoost models. For each component, the mean feature importance and standard deviation across the ensemble models were calculated to summarize its relative contribution to predicted OD600.

To further examine whether the effects of individual medium components depended on their combinations with other components, pairwise interaction effects were evaluated using linear regression models containing interaction terms. A baseline multiple linear regression model was first fitted using the concentrations of peptone, meat extract, yeast extract, and glucose as independent variables and OD600 as the dependent variable. Model performance was assessed using the Pearson correlation coefficient (r) between the observed and fitted OD600 values.

For each pair of medium components, an additional product term representing their interaction was added to the baseline model. The Pearson correlation coefficient was recalculated after inclusion of each interaction term, and the change in model performance (Δr) was calculated as the difference between the Pearson r of the interaction model and that of the baseline model. Larger positive Δr values were interpreted as indicating that inclusion of the corresponding pairwise interaction improved the model fit and that the effect of one component may depend on the concentration of the other. Pairwise interaction effects were visualized using a heatmap.

Statistical analysis

Statistical analyses of microbial community data were performed using non-parametric methods because microbiome abundance and diversity data were not assumed to follow a normal distribution. Differences in alpha-diversity indices between roadside and park soil

samples were evaluated using the Mann–Whitney U test. Beta-diversity differences in overall bacterial community composition were assessed using permutational multivariate analysis of variance (PERMANOVA) based on the corresponding distance matrices. A two-sided *P* value < 0.05 was considered statistically significant.

Taxonomic composition was examined descriptively across taxonomic levels. Differentially represented bacterial taxa were additionally explored using linear discriminant analysis effect size (LEfSe) at the order level. For LEfSe analysis, an FDR-adjusted *P* value < 0.10 and a log LDA score ≥ 2.0 were used as the predefined criteria.

The additional analysis of samples selected on the basis of contrasting microbiome profiles was treated as a post hoc exploratory analysis. Statistical results obtained from this subset were therefore interpreted descriptively and were not considered independent confirmatory evidence of differences between roadside and park environments.

For machine learning-guided culture optimization, experimentally measured OD600 values were used as the response variable. Feature importance values were summarized as the mean $\pm$ standard deviation across the 20 highest-performing XGBoost models. Pairwise interaction effects among medium components were evaluated descriptively by comparing Pearson correlation coefficients between baseline linear regression models and models containing individual interaction terms. The resulting change in correlation coefficient (Δr) was used as an indicator of improvement in model fit rather than as a formal statistical test of interaction. Model-predicted candidate medium compositions were interpreted as exploratory predictions unless experimentally validated.

Results

Urban soil sampling across roadside and park environments

A total of 32 urban soil samples were collected from roadside and park environments across the Seoul–Gyeonggi metropolitan region of the Republic of Korea. Roadside samples were obtained from sites located near major traffic routes, whereas park samples were collected from urban green spaces with minimal direct exposure to vehicular activity (Figure 1). Sampling sites were distributed across multiple districts in Seoul and Gyeonggi Province to capture spatial variation in urban soil environments.

Overall bacterial diversity and community composition in roadside and park soils

Bacterial community diversity was first compared between the complete roadside and park soil datasets. Alpha diversity, assessed using the Observed, Chao1, and Shannon indices, showed no statistically significant differences between roadside and park soils (Figure 2A). Similar patterns were observed for the additional alpha-diversity indices analyzed, indicating that overall bacterial richness and diversity were broadly comparable between the two land-use categories.

Beta-diversity analysis based on Bray–Curtis distance, Jensen–Shannon divergence, and Jaccard distance revealed substantial overlap between roadside and park soil samples in principal coordinates analysis (PCoA) space (Figure 2B). Although variation among individual samples was evident, the two groups did not form clearly separated clusters, suggesting considerable within-group heterogeneity in bacterial community structure.

At the phylum level, both roadside and park soils were dominated by several major bacterial phyla, with broadly similar overall community profiles between the two groups (Figure 2C). Differences in the relative abundance of individual phyla were observed across samples; however, the overall taxonomic composition showed substantial overlap between the two urban

soil environments.

Differential bacterial taxa between roadside and park soils

To identify bacterial taxa contributing to differences between roadside and park soils, LEfSe analysis was performed at the order level. Several bacterial orders exhibited relatively high LDA scores, indicating differences in their discriminatory contributions between the two soil environments (Figure 3).

Among the bacterial orders identified, Microtrichales satisfied the predefined LEfSe criteria, with an FDR-adjusted value of 0.083 and an LDA score of 3.8. This taxon therefore represented the most statistically supported differential bacterial order between roadside and park soils under the predefined exploratory significance threshold. Other bacterial orders exhibited comparatively high LDA scores but did not satisfy the corresponding FDR-adjusted significance criterion.

Post hoc exploratory analysis of microbiologically contrasting soil subsets

Because considerable heterogeneity was observed within the complete roadside and park datasets, a post hoc exploratory subset analysis was conducted using samples showing contrasting microbial community profiles in the initial NGS analysis.

Within the selected subsets, bacterial richness showed more pronounced differences than those observed in the complete dataset. The Observed and Chao1 indices differed between the selected roadside and park subsets, whereas the Shannon index showed no significant difference (Figure 4A). These findings suggest that the observed differences within the selected subsets were primarily associated with bacterial richness rather than overall diversity incorporating both richness and evenness.

PCoA based on Bray–Curtis, Jensen–Shannon, and Jaccard distances showed more visually

pronounced differences between the selected subsets than in the complete dataset, although substantial variability among individual samples remained (Figure 4B). Order-level taxonomic profiles also showed compositional differences between the selected subsets (Figure 4C). Because the subsets were selected retrospectively on the basis of the original microbiome profiles, these observations were considered descriptive and hypothesis-generating rather than independent evidence of differences between roadside and park environments.

Isolation and identification of culturable bacteria from urban soils

Culture-based isolation yielded a total of eight bacterial isolates from the collected soil samples, including three isolates recovered using MRS agar and five isolates recovered using nutrient agar. The isolates were purified based on colony morphology and subsequently identified by 16S rRNA gene sequencing followed by sequence comparison with the NCBI nucleotide database.

The isolates recovered on MRS agar were predominantly related to *Bacillus* and *Priestia*, with sequence identities generally ranging from 97% to 100%. In contrast, nutrient agar supported recovery of a broader range of culturable bacteria, including members of the genera *Bacillus*, *Paenibacillus*, and *Mycoplana*. Overall, the use of the two culture media enabled complementary recovery of culturable bacterial taxa from urban soil samples.

Among the recovered isolates, *Priestia megaterium* showed 100% sequence identity with the corresponding NCBI reference sequence and was selected as a representative soil-derived isolate for subsequent culture-medium optimization experiments.

Machine learning-guided optimization of *Priestia megaterium* culture medium

Growth of *P. megaterium* was initially evaluated across 12 manually designed medium compositions containing different concentrations of peptone, meat extract, yeast extract, and

glucose. The resulting OD600 measurements demonstrated variation in bacterial growth across the tested formulations and were subsequently used as the starting dataset for machine learning-guided medium optimization.

Following METIS-guided selection of a second experimental set, the overall distribution of measured OD600 values shifted upward relative to that observed in the first experimental round (Figure 5A). Examination of the individual medium components showed nonlinear and component-dependent relationships between nutrient concentration and bacterial growth (Figure 5B), indicating that no single component exhibited a simple monotonic relationship with OD600 across the tested concentration space.

Feature importance analysis using the XGBoost ensemble model identified **meat extract as the strongest predictor of bacterial growth (0.360 ± 0.076)**, followed by **yeast extract (0.276 ± 0.049)**, **glucose (0.253 ± 0.081)**, and **peptone (0.111 ± 0.114)** (Figure 5C). These results indicate that meat extract contributed most strongly to model predictions within the experimentally evaluated composition space.

Pairwise interaction analysis further showed that peptone, despite exhibiting the lowest individual feature importance, produced the largest improvements in model fit when combined with other medium components. The strongest effects were observed for peptone $\times$ meat extract ($\Delta r = 0.146$), peptone $\times$ glucose ($\Delta r = 0.143$), and peptone $\times$ yeast extract ($\Delta r = 0.102$) (Figure 5D). These patterns suggest that the contribution of peptone to bacterial growth may be highly dependent on the surrounding medium composition rather than acting independently.

Using the trained ensemble model, 5,000 feasible candidate formulations were subsequently evaluated *in silico*. High-predicted-growth compositions converged toward a region characterized by 5 g/L peptone, approximately 10 g/L meat extract, relatively low yeast extract concentrations, and 60 g/L glucose. Active learning further generated four candidate formulations containing 5 g/L peptone, 9.60–10.88 g/L meat extract, 0.64–3.12 g/L yeast

433 extract, and 60 g/L glucose for potential subsequent experimental evaluation. Because these
434 final model-recommended compositions were not tested in an additional cultivation round, they
435 were considered predicted candidate formulations rather than experimentally validated optimal
436 media.

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Discussion

In this study, we integrated 16S rRNA gene-based profiling of urban soil bacterial communities with culture-based bacterial isolation and machine learning-guided medium optimization. The comparison of roadside and park soils collected across the Seoul–Gyeonggi metropolitan region revealed substantial overlap in bacterial diversity and community composition, suggesting that broad land-use classification alone may not fully explain microbial variation across urban soils. Nevertheless, order-level analysis identified a limited differential signal, while a post hoc analysis of microbiologically contrasting subsets revealed more pronounced patterns in bacterial richness and community composition. In parallel, culturable soil bacteria were isolated and identified, and *Priestia megaterium* was subsequently used as a representative soil-derived isolate for machine learning-guided culture-medium optimization. Together, these results highlight both the substantial heterogeneity of urban soil microbiomes and the potential utility of integrating community profiling with data-driven cultivation approaches.

The overall similarity in alpha diversity between roadside and park soils indicates that bacterial richness and diversity were not strongly determined by land-use category within the present sampling framework. Likewise, the considerable overlap observed in beta-diversity ordinations suggests substantial variation among individual sampling sites within each environmental group. Urban soil bacterial communities are known to respond to multiple interacting environmental factors, including vegetation, soil physicochemical characteristics, moisture, organic matter, and anthropogenic disturbance [4,5,21]. Therefore, the absence of strong overall separation between roadside and park soils does not necessarily indicate that traffic-related or land-use-associated environmental factors have no influence on microbial communities. Rather, it suggests that the binary classification of sites as roadside or park may not capture the full environmental complexity underlying microbial community assembly.

Despite the broad overlap between the two groups, LEfSe analysis at the order level identified Microtrichales as the most statistically supported discriminative taxon under the predefined exploratory threshold, with an FDR-adjusted value of 0.083 and an LDA score of 3.8. This finding suggests that specific bacterial lineages may respond to differences between roadside and park environments even when community-wide diversity patterns remain relatively similar. However, because the statistical evidence was modest and the FDR threshold used for this exploratory analysis was 0.10, the association should be interpreted cautiously. Further studies incorporating direct measurements of soil properties and environmental contaminants will be necessary to determine whether variation in Microtrichales is reproducibly associated with specific characteristics of urban roadside environments rather than with unmeasured site-specific factors.

The post hoc analysis of microbiologically contrasting subsets provided additional insight into the considerable heterogeneity present within the complete dataset. In the selected subsets, differences in bacterial richness and taxonomic composition appeared more pronounced than in the full roadside-versus-park comparison. However, these samples were selected on the basis of patterns already observed in the original microbiome dataset, and the resulting analysis therefore cannot be regarded as an independent validation of environmental group differences. This analysis is more appropriately considered hypothesis-generating and illustrates the possibility that distinct microbial configurations may exist within broadly defined urban land-use categories. Future investigations using independently defined environmental criteria, such as concentrations of heavy metals or organic pollutants, soil pH, moisture, organic matter content, or vegetation characteristics, would be necessary to determine whether such microbiome patterns correspond to measurable environmental gradients.

In addition to culture-independent microbiome profiling, the present study incorporated culture-based isolation to obtain viable bacterial resources from urban soils. This

complementary approach is important because 16S rRNA amplicon sequencing provides information on community composition but does not directly provide viable isolates for subsequent experimental manipulation. In the present study, bacterial isolates recovered using MRS and nutrient agar represented several culturable soil-associated taxa, and *P. megaterium* was selected as a representative isolate for subsequent medium optimization. The use of both culture-independent and culture-dependent approaches therefore enabled the study to extend beyond descriptive community profiling toward experimental manipulation of a soil-derived bacterial isolate. Nevertheless, cultivation on a limited number of media inevitably captures only a fraction of the viable bacterial diversity present in soil, and the recovered isolates should not be considered representative of the complete soil bacterial community.

A major methodological component of this study was the application of a METIS-based active machine-learning framework to culture-medium optimization [12]. Following experimental screening of different combinations of peptone, meat extract, yeast extract, and glucose, the second experimental round showed an overall upward shift in measured bacterial growth compared with the initial screening round. The XGBoost ensemble model identified meat extract as the strongest predictor of OD600, followed by yeast extract, glucose, and peptone. Importantly, feature importance reflects the contribution of individual variables to model prediction and should not be interpreted as evidence that increasing a particular nutrient directly increases bacterial growth. The observed relationships between individual medium components and OD600 were non-linear, emphasizing the importance of considering multivariable composition rather than optimizing each nutrient independently.

Interestingly, although peptone showed the lowest individual feature importance, pairwise interaction analysis indicated that the largest improvements in model fit occurred when peptone interactions with meat extract, glucose, or yeast extract were included. This observation suggests that a component with relatively low individual predictive importance may still

influence growth in a composition-dependent manner. Such findings illustrate one potential advantage of data-driven optimization over simple one-factor-at-a-time approaches, as machine-learning frameworks can assist in exploring relationships within multidimensional experimental spaces [12]. Nevertheless, the interaction analysis performed here was descriptive: Δr represented improvement in model fit following inclusion of interaction terms rather than a formal statistical test of biological interaction. These patterns should therefore be regarded as candidates for future experimental validation.

The model further identified candidate regions of the medium-composition space predicted to support relatively high bacterial growth and generated additional formulations through active learning. However, these final model-recommended formulations were not evaluated in a subsequent cultivation experiment. Accordingly, the present findings demonstrate the feasibility of machine learning-guided candidate selection, rather than experimental confirmation of an optimal culture medium. This distinction is particularly important given the relatively small experimental dataset used for model development. Although ensemble modeling and cross-validation were employed to reduce model instability, a larger iterative dataset would be required to establish robust predictive performance and determine whether model-guided recommendations consistently improve growth across independent experiments. Several additional limitations should be considered. First, the study included a relatively limited number of soil samples collected within a single metropolitan region and during a single summer sampling period, limiting the ability to assess seasonal and geographic variability. Second, detailed soil physicochemical characteristics and direct measurements of traffic-related contaminants were not available. Therefore, differences observed between roadside and park soils cannot be attributed specifically to pollution, vehicle emissions, or any individual environmental factor. Third, 16S rRNA gene amplicon sequencing provides taxonomic information but has limited resolution for species-level identification and does not directly

538 establish microbial functional capacity. Fourth, the post hoc subset analysis was derived from
539 the original microbiome profiles and should therefore remain exploratory rather than
540 confirmatory. Finally, the culture optimization dataset was limited in size, and the final model-
541 recommended medium compositions were not experimentally validated. These limitations
542 emphasize the need for future studies combining longitudinal or independently sampled
543 cohorts, detailed environmental measurements, functional characterization of isolates, and
544 iterative experimental validation of machine learning-generated conditions.

545 Despite these limitations, the present study provides an integrated exploratory framework
546 connecting urban soil bacterial community characterization with culture-based isolation and
547 machine learning-guided cultivation. Rather than demonstrating a simple dichotomy between
548 roadside and park soil microbiomes, our findings highlight substantial microbial heterogeneity
549 across urban environments and suggest that more detailed environmental variables may be
550 required to explain community-level variation. At the same time, the application of active
551 machine learning to a soil-derived bacterial isolate illustrates a potential strategy for narrowing
552 complex culture-medium search spaces and prioritizing candidate experimental conditions [12].
553 Future validation and expansion of this approach may facilitate more efficient cultivation and
554 functional investigation of environmental microbial resources.

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Conclusion

This study characterized bacterial communities in roadside and park soils from the Seoul–Gyeonggi metropolitan region and integrated microbiome profiling with culture-based bacterial isolation and machine learning-guided medium optimization. Although the overall bacterial diversity and community structure showed substantial overlap between roadside and park soils, limited taxonomic differences were observed, including an exploratory order-level signal for Microtrichales. In addition, a post hoc subset analysis suggested that more distinct microbial patterns may exist within selected urban soil samples, although these findings should be interpreted as exploratory because the subsets were defined using the original microbiome profiles.

Culture-based isolation enabled the recovery of viable soil-derived bacterial strains, and *Priestia megaterium* was selected for subsequent culture-medium optimization. Machine learning-guided analysis identified meat extract as the most influential predictor of bacterial growth and revealed composition-dependent interaction patterns involving peptone. The model further generated candidate medium formulations predicted to support improved growth, although these final recommendations were not experimentally validated in an additional cultivation round.

Overall, this study demonstrates the feasibility of combining urban soil microbiome profiling, bacterial isolation, and active machine learning within a single exploratory framework. Future studies incorporating direct environmental measurements, larger independent sample sets, functional characterization of isolates, and experimental validation of model-recommended culture conditions will be required to confirm and extend these findings.

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Author contributions

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Consent for publication

Not applicable.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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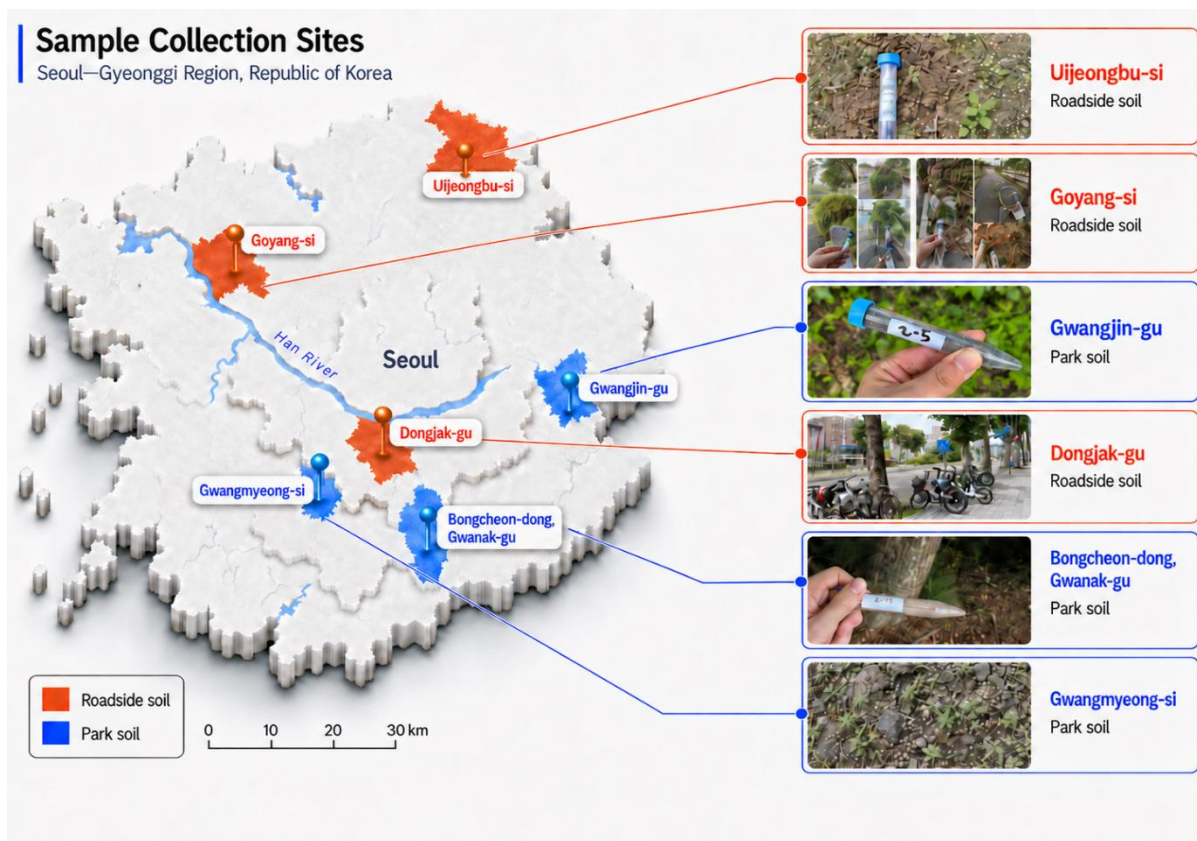


Figure 1. Sampling locations of urban soil samples in the Seoul–Gyeonggi metropolitan region, Republic of Korea.

A total of 32 soil samples were collected from roadside and park environments across Seoul and Gyeonggi Province. Roadside sampling sites are indicated in orange/red and included Uijeongbu-si, Goyang-si, and Dongjak-gu, whereas park sampling sites are indicated in blue and included Gwangjin-gu, Bongcheon-dong (Gwanak-gu), and Gwangmyeong-si. Representative photographs of the sampling environments are shown on the right. Roadside samples were collected from areas directly exposed to vehicular activity, whereas park samples were obtained from urban green spaces with minimal direct vehicle exposure.

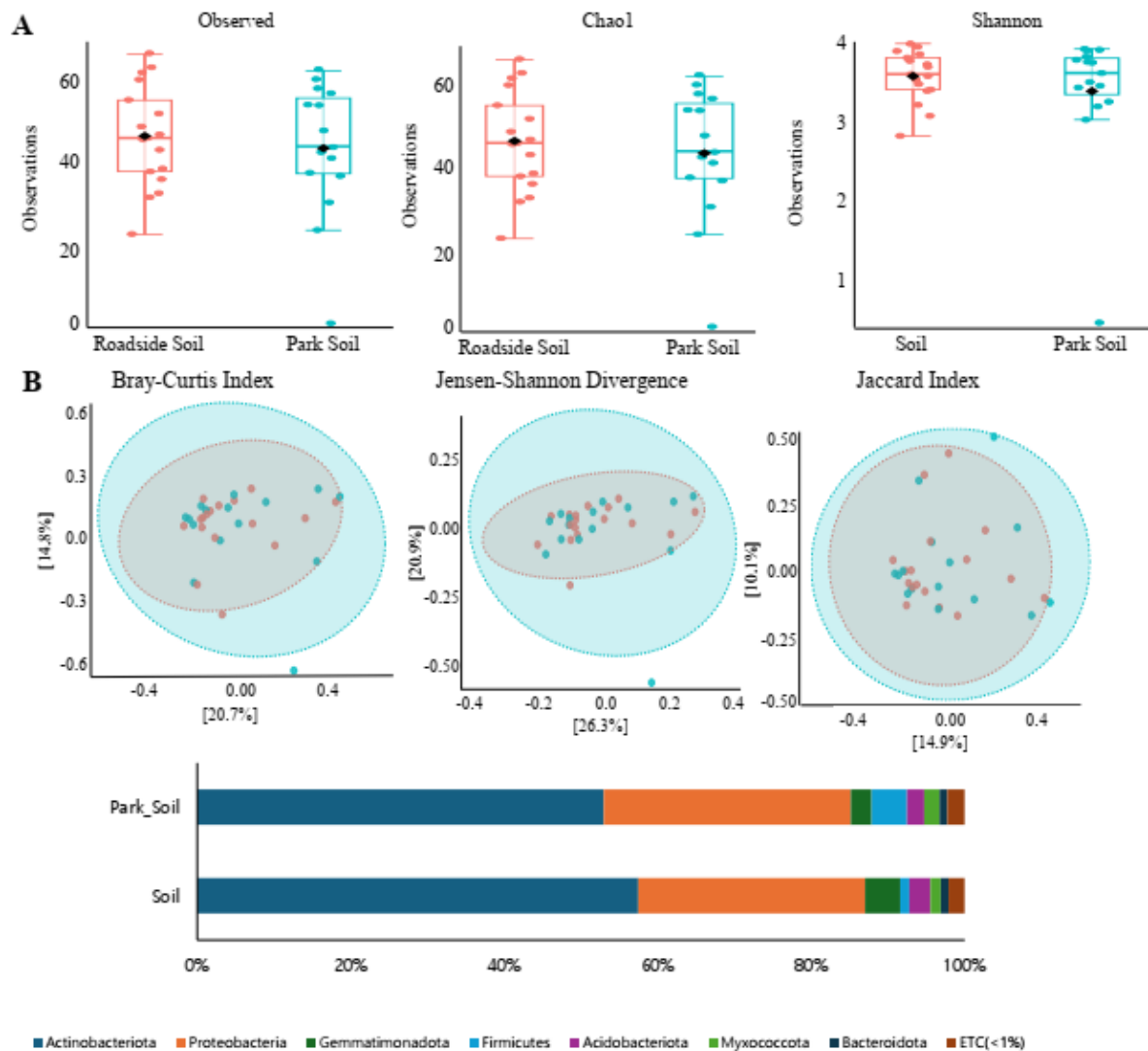


Figure 2. Bacterial diversity and taxonomic composition of roadside and park soil microbiomes.

(A) Alpha diversity of bacterial communities in roadside and park soils assessed using the Observed, Chao1, and Shannon indices. Box plots represent the distribution of diversity values across samples, with individual samples shown as points.

(B) Principal coordinates analysis (PCoA) of beta diversity based on Bray–Curtis distance, Jensen–Shannon divergence, and Jaccard distance. Each point represents an individual soil sample, and shaded ellipses indicate the distribution of samples within each group.

(C) Relative abundance of bacterial communities at the phylum level in roadside and park soils. Taxa with a relative abundance below 1% were grouped as ETC (<1%).

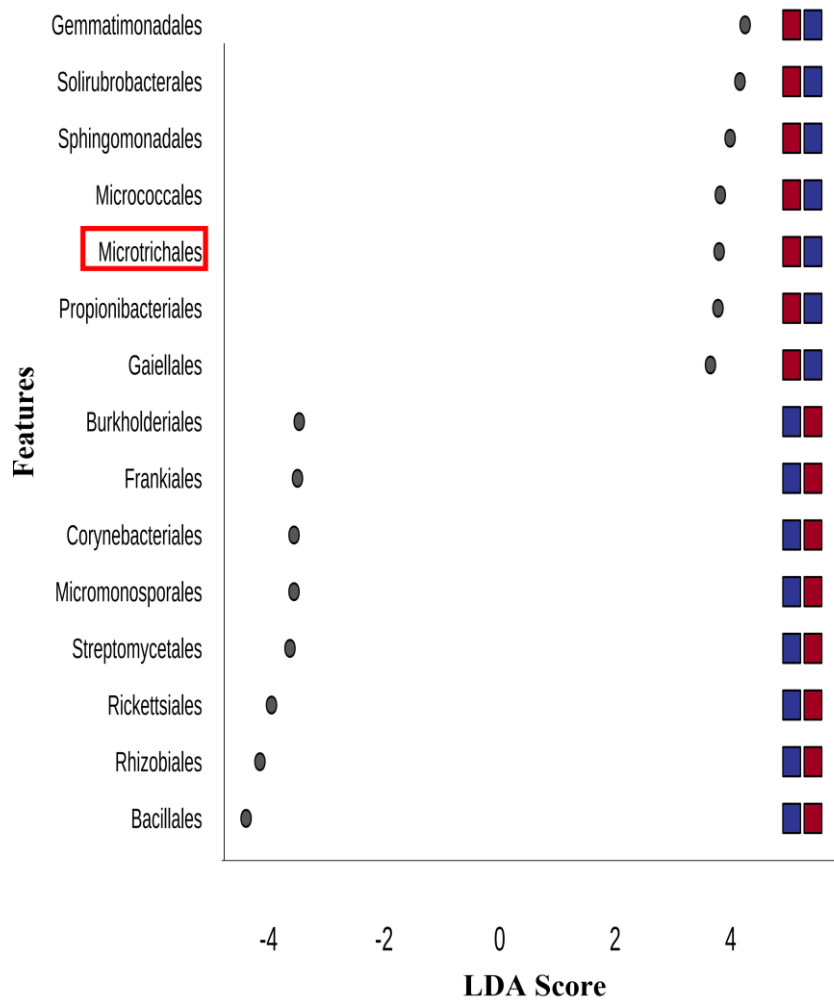


Figure 3. Linear discriminant analysis effect size (LEfSe) analysis of bacterial communities at the order level between roadside and park soils. LEfSe was performed using an FDR-adjusted *P*-value cutoff of 0.10 and a log LDA score threshold of 2.0. Taxa are displayed according to their LDA scores, with positive and negative values indicating enrichment in the respective soil groups. *Microtrichales*, highlighted by the red box, satisfied the predefined FDR-adjusted significance criterion (FDR = 0.083; LDA score = 3.8).

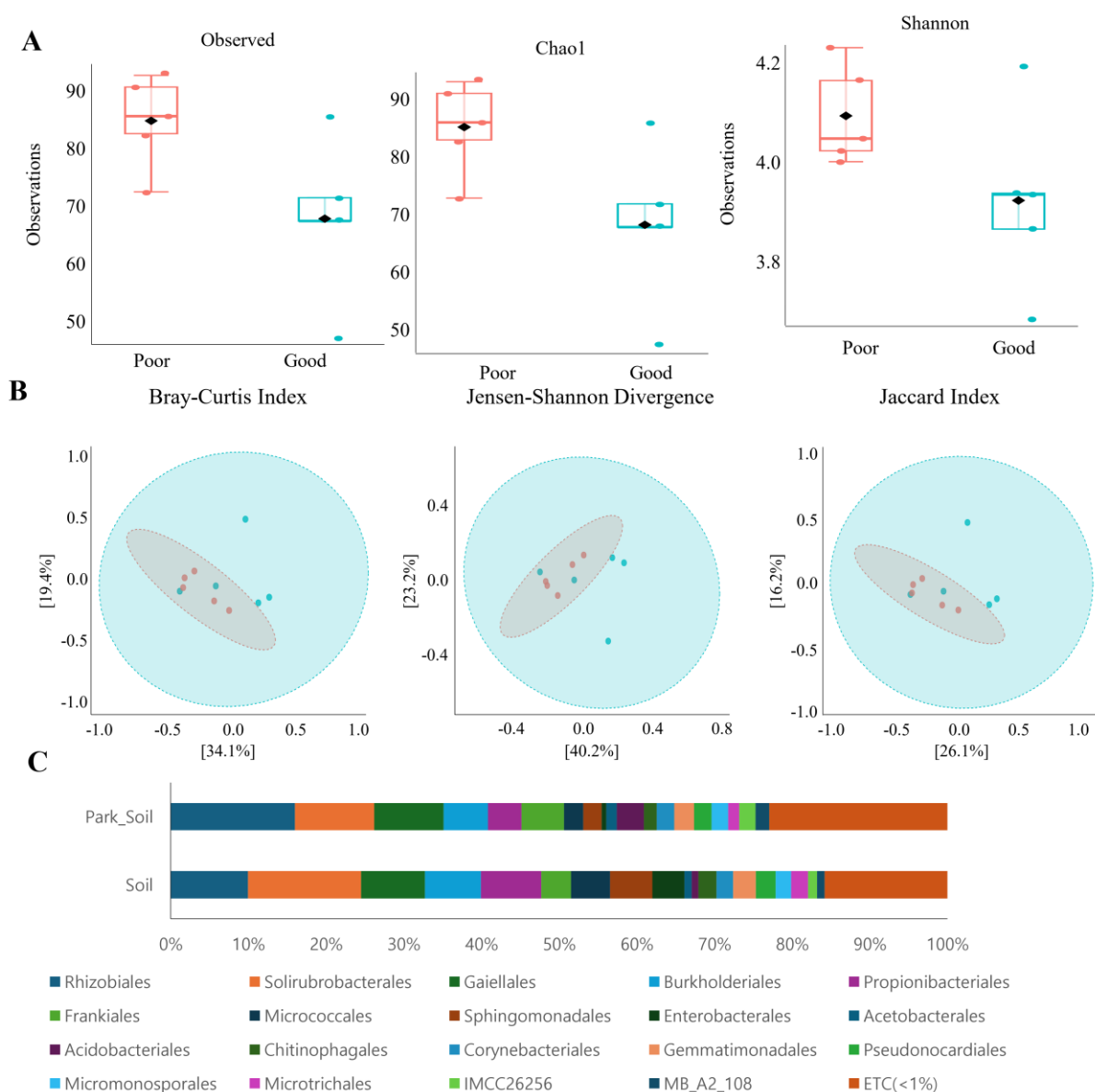


Figure 4. Post hoc exploratory analysis of microbiologically contrasting roadside and park soil subsets.

(A) Alpha diversity of the selected soil subsets assessed using the Observed, Chao1, and Shannon indices. Box plots show the distribution of diversity values for each group, with individual samples represented as points.

(B) Principal coordinates analysis (PCoA) of beta diversity based on Bray–Curtis distance, Jensen–Shannon divergence, and Jaccard distance. Each point represents an individual soil sample, and shaded ellipses indicate the distribution of samples within each group.

(C) Relative abundance of bacterial communities at the order level in the selected roadside and park soil subsets. Taxa with a relative abundance below 1% were grouped as ETC (<1%). The subsets were selected post hoc based on contrasting microbial community patterns observed in

720 the initial NGS analysis; therefore, these findings were interpreted as exploratory rather than
721 as independent validation.

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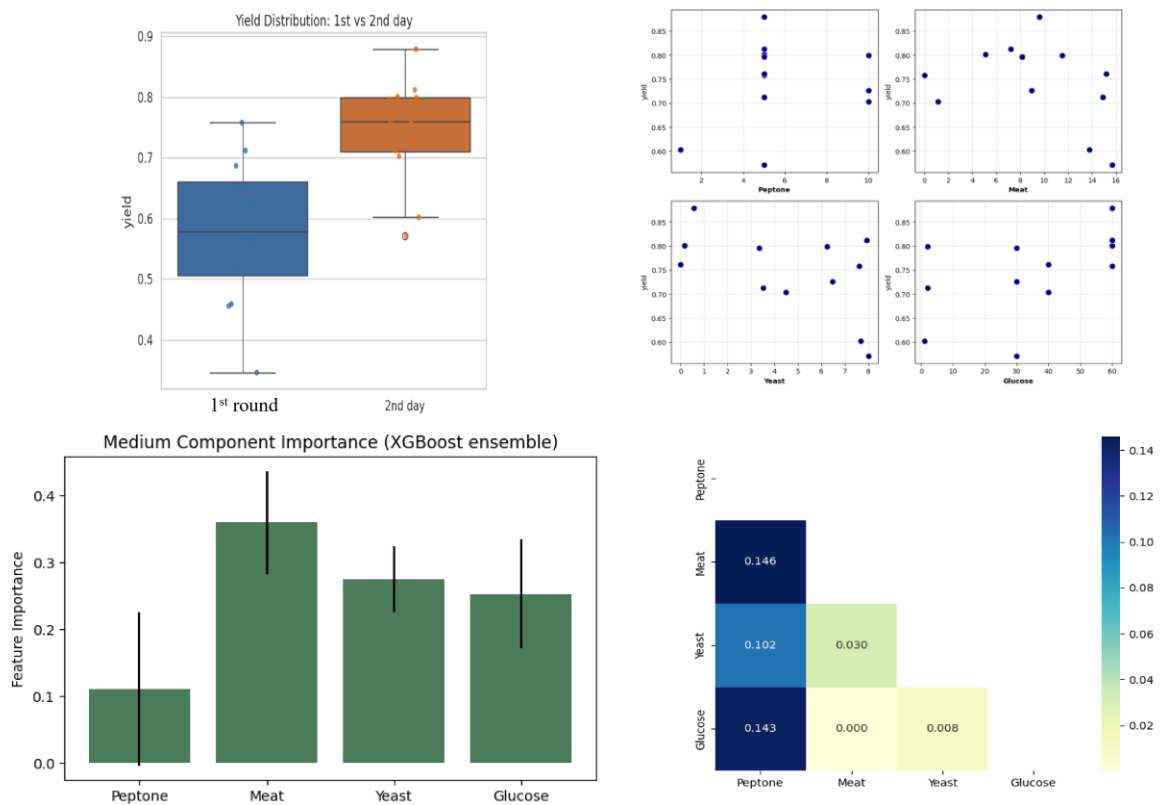


Figure 5. Machine learning-guided culture-medium optimization for *Priestia megaterium*.

Machine learning-guided culture-medium optimization for *Priestia megaterium*. (A) Box-and-whisker plots comparing the distribution of bacterial growth yields (OD600) obtained in the first and second experimental rounds of medium optimization. The second round showed an overall upward shift in yield distribution compared with the first round. (B) Scatter plots showing the relationships between the concentrations of peptone, meat extract, yeast extract, and glucose and the corresponding bacterial growth yields obtained in the second round. Each point represents an individual medium composition. (C) Relative feature importance of the four medium components estimated using the XGBoost ensemble model. Meat extract showed the highest feature importance, followed by yeast extract, glucose, and peptone. Error bars indicate the standard deviation across ensemble models. (D) Pairwise interaction heatmap showing improvements in model fit (Δr) after inclusion of individual interaction terms in linear regression models. The strongest interaction effects were observed for peptone with meat

738 extract, glucose, and yeast extract, suggesting that the contribution of peptone to bacterial
739 growth may depend on its combination with other medium components.
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